Customer Churn: Predictive Analytics for Telecom Industry

Abstract. In the highly competitive telecommunications industry, customers are the most important source of livelihood for telecommunications providers. Telecommunications companies place more emphasis on customer retention than on customer acquisition, as acquiring new customers involves considerable costs. The churn of long-standing customers leads to a decline in the market shares of telecommunications providers. To tackle the problem of customer churn, it is critical for business analysts specialists to have a comprehensive understanding of the behavioural patterns inherent in existing customer churn data.This study uses three unique models, including LGBM, logistic regression and random forest, to develop a reliable and accurate model for predicting customer churn specifically for the telecommunications sector. Moreover, the results of the empirical evaluation clearly show that logistic regression, which was selected based on the AUC measure, is the most appropriate model for this particular objective. The aforementioned findings provide important and necessary insights into the characteristics and preferences of customers. This information enables the prediction of potential churn, the identification of reasons for customer churn and the implementation of proactive steps to avoid churn.

Keywords: Customer Churn · Logistic Regression · telecom

1. Introduction

Telecommunications companies face significant obstacles in attracting new customers, and the financial impact of customer churn is severe. In the telecommunications sector, it is well known that the cost-effectiveness of maintaining existing customers is higher than that of acquiring new ones. Essentially, the customer base a company already has is of greater value and an easier target to sell its products to, as opposed to a company that is constantly looking for new customers. An established customer base has the advantage of acting as a brand advocate, often promoting products through word-of-mouth recommendations.

In addition, customer loyalty brings several benefits, including evaluating a company's performance in emerging markets, improving return on investment (ROI) and indicating the level of service quality delivered. The factors of customer satisfaction, trust and commitment are of great importance in the context of customer loyalty in a company. Nevertheless, achieving complete customer retention, especially in the telephone industry, can be difficult due to strong competition.

The phenomenon of customer loss within a company is sometimes referred to as churn rate, customer churn rate or simply churn. This ratio indicates the proportion of customers who end their business relationship with a company. High churn rates in the telecommunications industry are sometimes the result of inadequate customer service, poor service delivery and unresolved issues. These elements contribute to significant challenges that lead to financial losses and, in certain cases, damage the company's reputation.

The telecom industry has a long-standing issue of consistently losing customers, leading to a reduction in their revenues and negative impacts on their overall business. Telecom companies offer a range of services, including cell phone services, Internet, online services, technical support, and various subscription-based offerings. As a result, these companies are actively seeking ways to improve their customer retention programs. To address this challenge, organizations require predictive models capable of forecasting customer churn. These models are essential for implementing strategies that can effectively enhance customer retention. In essence, organizations and their leaders are actively seeking predictive models to combat customer churn by implementing methods aimed at boosting customer retention rates.

Previous research indicates that customer churn reduction is critical in reducing organizational losses compared to chasing new customers (T. Vafeiadis et. Al (2015)). According to statistics, the average monthly customer churn rate for Telco companies in the United States stands at 1.9% when considering the four major companies. However, when prepaid services are factored in, the churn rate is projected to increase to 67% annually. Research underscores the severity of the customer churn issue in the telecom industry, with annual churn rates ranging from 32% to 35%. This high churn rate necessitates that companies aim for higher retention rates. To put this into perspective, in an industry where customer churn reaches as high as 35%, companies must maintain a customer retention rate of at least 68% to preserve a significant market share. This highlights the importance of implementing effective customer retention strategies in the face of considerable customer attrition. (A. Moreno (2022)).

The data presented clearly indicates that increased customer turnover rates have a significant negative impact on a company's turnover and overall success in the industry. Previous studies have shown that building reliable predictive models for customer turnover requires the use of a simple value model to better understand the problem. This approach facilitates the development of more effective customer retention strategies for companies, leading to an overall improvement in customer satisfaction. By understanding the many elements that contribute to customer attrition and accurately predicting it, companies can adopt proactive strategies to limit customer churn and cultivate a more satisfied and loyal customer base. (V. V. Saradhi, G. K. Palshikar, 2012).

The utilization of the value model revolves around the identification of customers who hold the highest value within an organization. This approach serves as a foundation for devising retention strategies aimed at preserving a larger number of these valuable customers. In more

extensive research, the authors emphasize the imperative for telecom companies to gain a comprehensive understanding of the factors that contribute to customer churn. Such insights are crucial for these companies as they enable them to take proactive measures and implement strategies designed to mitigate the problem of customer churn. By addressing the root causes of churn, telecom companies can work towards retaining a larger portion of their customer base and securing long-term success (P. Lalwan et. Al, 2022). Leveraging machine learning techniques on vast sets of big data is instrumental in creating predictive models for customer churn in these organizations. Previous research that employed machine learning for building such predictive models proposed using the Area Under Curve (AUC) metric, resulting in an impressive accuracy rate of 93.3%. In addition to AUC, this research incorporated features from Social Networking Analysis (SNA) to enhance the model's performance even further. The inclusion of SNA features likely contributed to the model's ability to make more accurate predictions regarding customer churn, underscoring the importance of incorporating various data sources and sophisticated techniques in churn prediction models.

There are various machine learning and visualization models that can be highly useful in customer churn prediction. One such example is HEAVY.AI, which is an accelerated analytic model designed to process extensive big data sets from customers. These advanced models can offer more accurate insights and predictions by analyzing large volumes of data, identifying patterns, and enabling organizations to proactively address customer churn. (K. A. Saran Kumar, D. Chandrakala,2016). Using models like HEAVY.AI, companies can gain a competitive edge by making data-driven decisions to improve customer retention and satisfaction. Such tools can be crucial in the telecom industry and other sectors where customer churn is a significant concern, allowing companies to implement targeted strategies to reduce churn and boost overall customer experience. From the studies, it appears that this model, HEAVY.AI, has gained the trust of well-known telecommunications companies. This trust enables these companies to effectively analyse customer churn using advanced analytics. The results of these analytics play a crucial role in detecting irregularities, managing the fleet of services or customers and improving overall reliability. By using such trusted models and analytics, these companies can not only mitigate the negative impact of customer churn, but also optimise their operations and services to provide a better customer experience and maintain their competitive advantage.

According to research, the churn prediction problem involved models that only predicted customer churn for the current or next month (A. Amin, et al., 2016). models that predict customer churn over a very short period may prove ineffective, as organizations often require adequate time to develop and implement strategies aimed at retaining those customers. In light of this, the study suggests the adoption of a new T + 2 churn customer prediction model, proposed by Z. Zhu (K. Dahiya, S. Bhatia, 2015). This model likely provides a longer lead time for organizations to take necessary actions in response to predicted churn, thus improving their chances of successful customer retention. Predictive models like the T + 2 churn customer prediction model allow companies to proactively address churn and develop tailored retention strategies, ultimately bolstering customer satisfaction and long-term business performance. The newly introduced T+2 model is characterised by its ability to predict customer churn within a

two-month time frame. This feature provides telecom companies with a great opportunity to plan, develop and execute customer retention initiatives. Using an extended forecasting window can provide significant benefits to these companies.

The main objective of this study is to investigate a customer churn prediction model developed specifically for telecommunication companies, focusing on the effective presentation of the model results. The objective of this research is achieved by applying a systematic methodology that includes many different sections:

Data and Methodology section: this part of the article contains a detailed presentation and explanation of the different forecasting models used. The text begins with an overview of the Light Gradient Boosting Machine (LGBM) model, followed by an examination of logistic regression and random forest models.

Results and Discussion: This section aims to present the results and findings of the study. This analysis examines the results of the customer churn prediction model and presents a comprehensive discussion that interprets these results, enabling the extraction of valuable insights and a deeper understanding of the ramifications involved. Identifying and explaining limitations or constraints in the study process is a crucial aspect to be addressed. This part plays a crucial role in providing a well-rounded and honest analysis of the results.

Finally, this study provides a summary of the main findings and the significance of the customer churn prediction model created for telecom companies. In addition, the importance of the two-month forecasting window offered by the T + 2 model can be highlighted.

This methodology guarantees a comprehensive evaluation of the customer churn prediction model, while maintaining transparency and taking into account any constraints. The use of this tool has the potential to help telecom companies make informed decisions to reduce churn and improve customer retention.

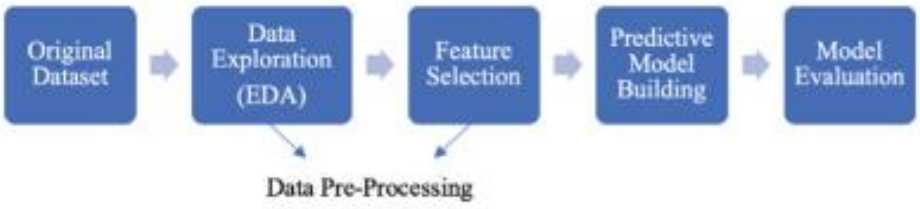


Figure 1: Step model for predicting the customer churn framework.

The graphical representation of the research architecture can be seen in Figure 1. The process consists of many stages, including data search, feature selection, model development and evaluation. The proposed paradigm includes a number of different stages. The process consists of five different steps, the first of which involves identifying the most relevant data. In this study, exploratory data analysis (EDA) was used to investigate the relationships between variables using analysis of variance and correlation matrix. In addition, missing data are accounted for through the estimation and deletion methods. Stage 2: Feature selection. Stage 3 involves the development of predictive models, in particular using the Light Gradient Boosting Machine (LGBM), Logistic Regression and Random Forest algorithms. Finally, the evaluation of the predictive models is carried out, using accuracy and AUC curves as evaluation measures.

2. Methodology

2.1 Exploratory data analysis (EDA)

Exploratory data analysis (EDA) involves the study of inherent characteristics within a data set through the use of visualisation, summarisation and analysis techniques. The main objective of EDA is to improve the understanding of the data set and thus facilitate the identification of missing values. Exploratory data analysis (EDA) is about identifying potential variables in a given database that could effectively explain or differentiate consumer behaviour. Exploratory data analysis (EDA) is able to perform bivariate analysis to identify and explore the association between two variables. Furthermore, it is able to identify relationships between multiple variables and thus eliminate redundant characteristics.

2.2 Light Gradient Boosting Machine

The Light Gradient Boosting Machine (LGBM) is a machine learning model known for its unique features, such as superior accuracy and faster training speed compared to other models. The system uses a machine learning algorithm optimised for efficient processing of data (G. Ke, et al, 2017) . The above features contribute to LGBM's use in many fields involving predictive modelling tasks that require optimal speed and accuracy to achieve the desired results. The model's ability to effectively manage large datasets and its efficiency in training make it a desirable tool for tasks such as predicting customer churn in the telecommunications sector .

In addition, the LGBM algorithm demonstrates horizontal scalability and helps identify areas where customer churn may occur. The LGBM method uses both discrete and continuous feature values. It applies a histogram algorithm to arrange discrete values into bins. This approach determines the best cut-point for the analysis of all available characteristics and data.

2.3 Logistics Regression

Logistic regression is a regularly used statistical modelling technique, usually employed to solve binary classification tasks (D. W. Hosmer, S. Lemeshow, 2018). Logistic regression is used to predict a dependent data variable through the analysis of the relationship between one or more existent independent variables. The model can be written as

$$P(Y = 1|X) = \frac{1}{1+e^{-\sum_{i=1}^{n} \beta_i X_i - \beta_0}} \quad (1)$$

where P(Y = 1|X)is the probability function predicting a positive outcome, y is a binary dependent variable representing whether things happen or not (if Y = 1, it means things happen; otherwise, Y = 0), β0 ~ βn describe the different regression coefficients, X1 ~ Xn are the independent value that affects the dependent variable. In addition, logistic regression is essentially sigmoid. The logistic regression results appear in binary form, i.e., "yes" or "no," rather than as numbers, making it easy to plot similar curves. The first step of logistic regression analysis involves preprocessing the data. Then, logistic regression uses the existing customer data to identify or predict upcoming sales. Ultimately, the process involves creating a confusion matrix, visualising the results from the test set and then analysing them.

2.4 Metrics

This section explains the evaluation measures used in this paper, including precision, recall, F1 score, accuracy, confusion matrix and AUC. The above measures are used to quantitatively evaluate the efficiency of a predictive model. First, the results of the empirical investigation are examined using the most demanding evaluation metrics, which include precision, recall and the F1 score. Precision assesses the proportion of correctly predicted churned customers within the predictions. Recall measures the probability of correctly identifying true churn within the entire customer base. The F1 score, which acts as a comprehensive metric, assesses classification performance. Accuracy is the ratio of all correctly predicted scores. In the context of binary classification, accuracy can also be assessed separately for positive and negative outcomes. The mathematical formulas of these evaluation metrics are presented below. More details can be obtained from the study mentioned in A. Amin, 2017.

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

$$F1score = \frac{2(Precision \times Recall)}{(Precision+Recall)} \quad (4)$$

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (5)$$

In this context, the abbreviation TP stands for True Positives, TN for True Negatives, FP for False Positives and FN for False Negatives.

The confusion matrix includes four variables that act as performance indicators and it is commonly used to illustrate the results. The matrix under consideration is applicable to the binary classification task, where the actual and predicted values are plotted on separate axes.

• True Positive (TP): Both predicted and actual are positive
• True Negative (TN): Both predicted and actual negative
• False Positive (FP): Predicted positive, actual negative
• False Negative (FN): Predicted negative, actual positive

Once these four variables are ascertained, it is possible to calculate the True-Positive Rate (TPR), False-Negative Rate (FNR), True-Negative Rate (TNR), and False-Positive Rate (FPR). In the context of a confusion matrix, the optimisation objective is to maximise the true positive rate (TPR) and true negative rate (TNR), while simultaneously minimising the false positive rate (FPR) and false negative rate (FNR).

$$TPR = \frac{TP}{ActualPositive} = \frac{TP}{TP+FN} \quad (6)$$

$$FNR = \frac{FN}{ActualPositive} = \frac{FN}{TP+FN} \quad (7)$$

$$TNR = \frac{TN}{ActualNegative} = \frac{TN}{TN+FP} \quad (8)$$

$$FPR = \frac{FP}{ActualNegative} = \frac{FP}{TN+FP} \quad (9)$$

In evaluating the predictive models, we used the area under the Receiver Operating Characteristic Curve (AUC), a critical metric for evaluating the model's ability to find the most effective high-performance model for the given data set. The area under the Receiver Operating Characteristic (ROC) curve, commonly referred to as AUC, is a widely used statistic for evaluating the performance of a probabilistic classifier (D. J. H, R. J. Till, 2001). The value of AUC represents the performance of the model. The perfect classifier will have a solid, valid positive rate and a low false-positive rate (AUC = 1). The formula for calculating AUC is below:

$$\hat{A} = \frac{S_0 - n_0(n_0+1)/2}{n_0 n_1} \quad (10)$$

where n0 means the number of positive examples and n1 means the number of negative examples. Also, S0 = ∑ri, where ri is the rank of the i th positive example in the ranked list.

3. Results

Tables 1, 2 and 3 provide a graphical representation of the evaluation criteria for the three models. The following tables show the Precision, Recall, F1 Score and Accuracy metrics for each model. The data in the tables are divided into two categories: "True" and "False" The label "True" represents the occurrence of churn, while the label "False" represents the absence of churn. In the "False" category, it can be seen that the logistic regression has the highest precision value, 0.93. The accuracy of the random forest model is comparatively lower, but its

recognition is the highest, at 0.92. The F1 values of the three models, namely 0.83, 0.81 and 0.87, show no statistically significant difference. In connection with the aspect "True", it can be observed that Random Forest has the best precision and at the same time the lowest recall.

The observed result is in contrast to the category "False", where the LGBMClassifier has the best precision, while logistic regression has the highest recall. Both models have comparable F1 values, with the Random Forest model achieving an F1 value of 0.54. The measure of performance that is easiest to understand and interpret is precision. The accuracy of the Random Forest model, as shown in Tables 1, 2 and 3, is 79%, the highest among the three models. The accuracy of the other two modifications is 77% and 75% respectively.

	Precision	Recall	F1-Score	Support
False	0.88	0.79	0.83	1697
True	0.56	0.72	0.63	628
Accuracy			0.77	2325
Macro avg	0.72	0.75	0.73	2325
Weighted avg	0.80	0.77	0.78	2325

Table 1: LGBM Classifier

	Precision	Recall	F1-Score	Support
False	0.93	0.72	0.81	1697
True	0.53	0.84	0.65	628
Accuracy			0.75	2325
Macro avg	0.73	0.78	0.73	2325
Weighted avg	0.82	0.75	0.77	2325

Table 2: Logistic Regression

	Precision	Recall	F1-Score	Support
False	0.82	0.92	0.87	1697
True	0.67	0.46	0.54	628
Accuracy			0.79	2325
Macro avg	0.75	0.69	0.70	2325
Weighted avg	0.78	0.79	0.78	2325

Table 3: LGBM Classifier

Utilizing the confusion matrix offers the advantage of assessing the model's performance, even in scenarios with imbalanced data, and permits the calculation of various other performance metrics. Figures 2, 3, and 4 clearly depict noteworthy differences in the True Negative Rate across the three models. Notably, the True Negative Rate of the Random Forest model stands out as the highest, at 0.92. However, it is important to note that the True Positive Rate (TPR) of the Random Forest model is comparatively lower than that of the other models, with Logistic Regression leading the way. Specifically, the Random Forest has a TPR of 0.46, while Logistic Regression boasts a TPR of 0.84.

Turning our attention to the "Negative" aspect, the Random Forest model demonstrates the lowest False Negative Rate (FNR) at 0.083, whereas Logistic Regression exhibits a higher FNR of 0.28. It is worth noting that the False Positive Rate (FPR) results are inversely proportional to the FNR outcomes. In this context, the Random Forest model shows the highest FPR among the three models, standing at 0.54. In contrast, the FPR for Logistic Regression is notably lower at only 0.16.

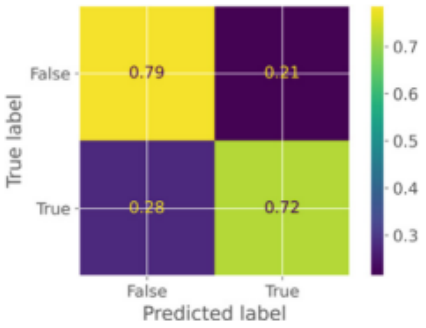


Figure 2: Confusion Matrix of LGBM Classifier

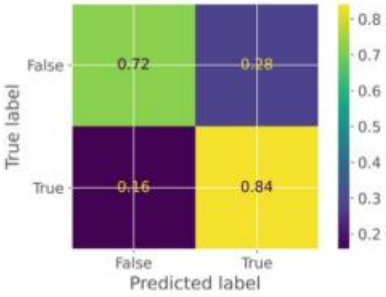


Figure 3: Confusion Matrix of Logistic Regression

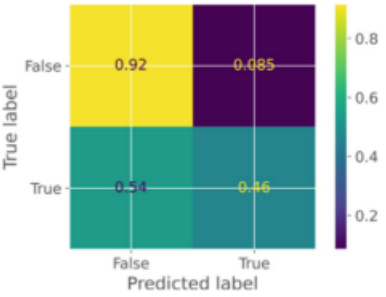


Figure 4: Confusion Matrix of Random Forest

Based on Figures 5, 6, and 7, it becomes evident that the AUC values for LGBM Classifier, Logistic Regression, and Random Forest are 75%, 78%, and 69%, respectively. Notably, the AUC value of Logistic Regression surpasses that of the other models, positioning it as the most effective classification algorithm for this prediction model. LGBM Classifier follows with the second-highest AUC value. In contrast, the AUC value of Random Forest ranks last, likely due to the data's inherent imbalance.

Two promising models were discovered in this study: random forest, which was evaluated based on its accuracy, and logistic regression, which was based on the AUC. However, when random client variables were used to evaluate the performance of both models, it became clear that the results for the Random Forest model were unbalanced, as shown in Figure 8. This discovery suggests that the Random Forest algorithm may not be suitable for all customer variables and may perform better in detecting the different characteristics of churned consumers.

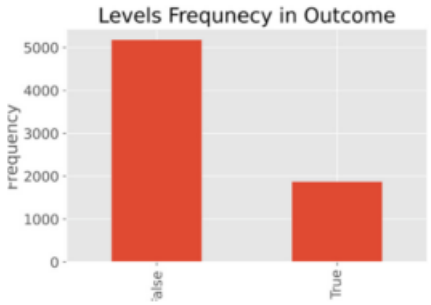


Figure 8: Random Forest Outcome with arbitrary customer feature.

This study analysed the results of numerous model tests and concluded that Random Forest was the most optimal predictive model in terms of accuracy and recall. On closer inspection, it became clear that the use of Random Forest led to a certain degree of imbalance in the data. This suggests that Random Forest is better suited for identifying and analysing the underlying factors that contribute to customer turnover. Although Random Forest is not very effective in accurately identifying churned customers, it has proven successful in identifying a broader group of customers. This coincides with one of the main objectives of customer churn management.

However, logistic regression has shown better predictive capabilities when evaluated against AUC curves, which are a measure of the model's performance compared to random guessing. Therefore, it is a viable option for predicting and identifying customers who are vulnerable to churn.

However, it is important to recognise that this particular model has inherent limitations, mainly related to the consumer characteristics on which it depends. Consumer characteristics can change as customers undergo evolutionary changes. Predicting customers with unique characteristics is challenging due to the limited scope of information, which only includes attributes that are prevalent among the majority of customers. To improve the accuracy of predicting customer churn, it is recommended that future studies use additional customer attributes, such as call type, complaint information and geographic data.

Furthermore, it is imperative to acknowledge the existence of unbalanced categorisation within this research. The uneven distribution of individuals may impact the selection of future

models. Future research efforts should prioritise addressing this issue of unbalanced categorisation.

4. Conclusion

In summary, this study examines several predictive classifiers, particularly their application in predicting customer turnover in the telecommunications industry. The telecommunication industry is witnessing a continuous evolution of services in line with technological advancements. Therefore, the ability of telecommunication providers to anticipate and effectively manage customer churn offers increasing benefits.

This research article uses various machine learning approaches to predict customer churn in the telecommunications industry. In particular, the Light Gradient Boosting Machine, Logistic Regression and Random Forest models are used. The research shows that Logistic Regression is the most appropriate model when evaluated against AUC metric, demonstrating its efficiency in accurately identifying customers at risk of churn. In contrast, Random Forest has exceptional performance in terms of accuracy, making it an excellent tool for analysing the variables that contribute to customer churn.

It is important to recognise that the work of predicting customer churn is complex. To improve the accuracy and precision of the predictions, it may be beneficial to add new customer variables to the database in the future.

In summary, our model shows promise in helping telecom providers correctly and effectively predict customer churn and match their offerings to consumer demand. The results reported in this scientific article provide a useful framework for predicting customer churn.

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